

Semester	T.E. Semester VI - CMPN
Subject	DevOps Lab
Subject Professor In-charge	Dr. Snehal Andhare
Assisting Teachers	Dr. Snehal Andhare

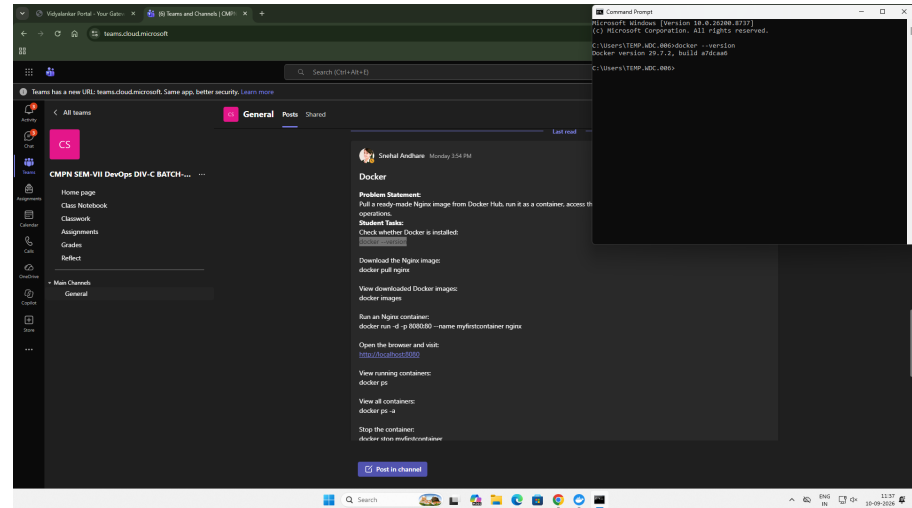
Laboratory

Student Name	Vijay Hanumandla
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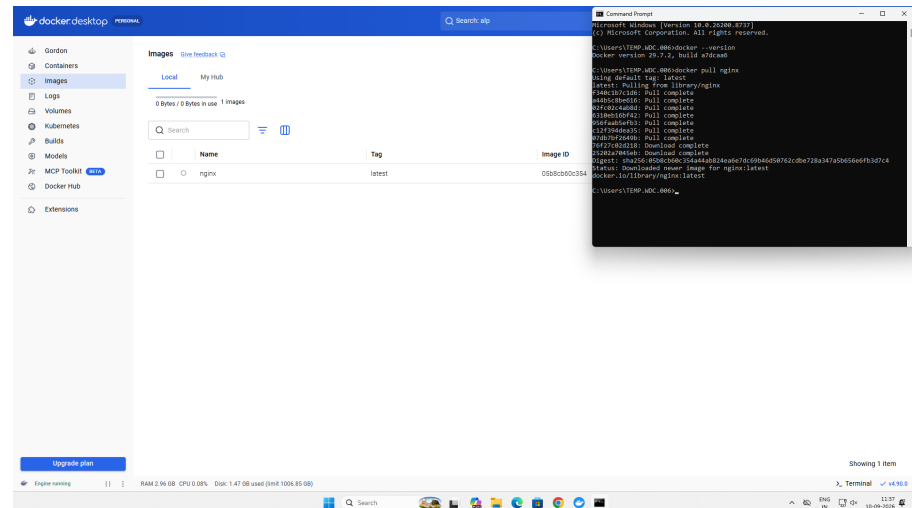
Experiment Number	07
Experiment Title	Docker Environment Verification and Nginx Container Operations
Objectives (Skill Set / Knowledge Tested / Imparted)	The primary objective of this experiment is to verify the Docker environment, work with Docker images, run an Nginx web container, access the service through a browser, and perform basic container lifecycle operations such as listing, stopping, starting, and removing containers.
Description	This lab introduces Docker through a sequence of practical tasks covering environment verification, image retrieval and inspection, Nginx container execution, browser access, container inspection and shell access, page modification, persistence, bind mounts, multiple web servers, port conflicts, logs, monitoring, networking, inter-container communication, lifecycle management, and cleanup. The attached execution evidence documents the Docker version check, Nginx image retrieval, container creation, browser access, container status checks, and lifecycle operations.
Program	Docker Environment Verification and Nginx Container Operations Key commands demonstrated: docker --version docker pull nginx docker images docker run -d -p 8080:80 --name myfirstcontainer nginx docker ps docker ps -a docker stop myfirstcontainer docker start myfirstcontainer docker rm myfirstcontainer

Part A / Part B

Task 1: Verify Docker Installation

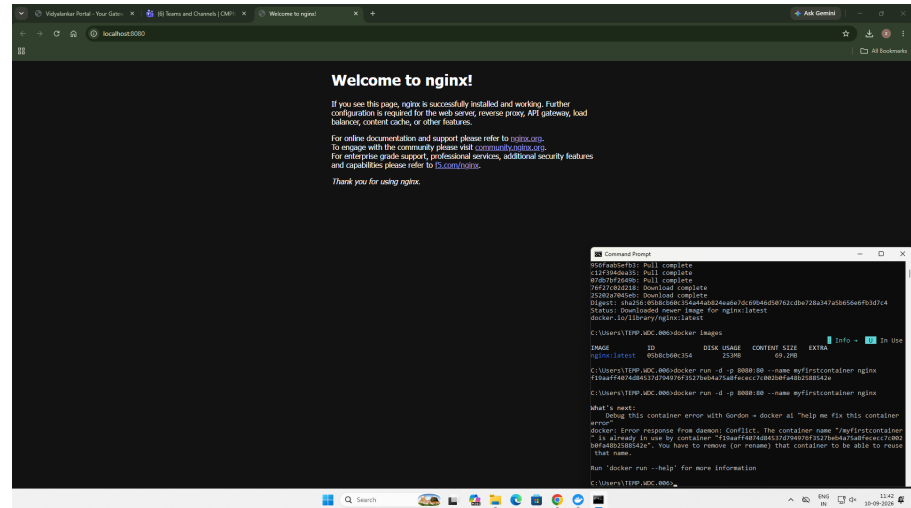


Task 2: Pull the Nginx Image

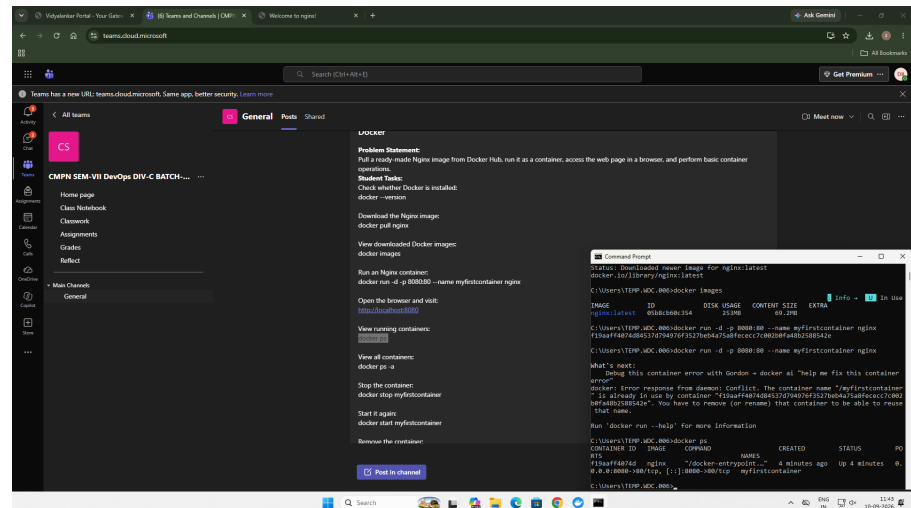


Container Access

Task 4: Open the Nginx Web Page

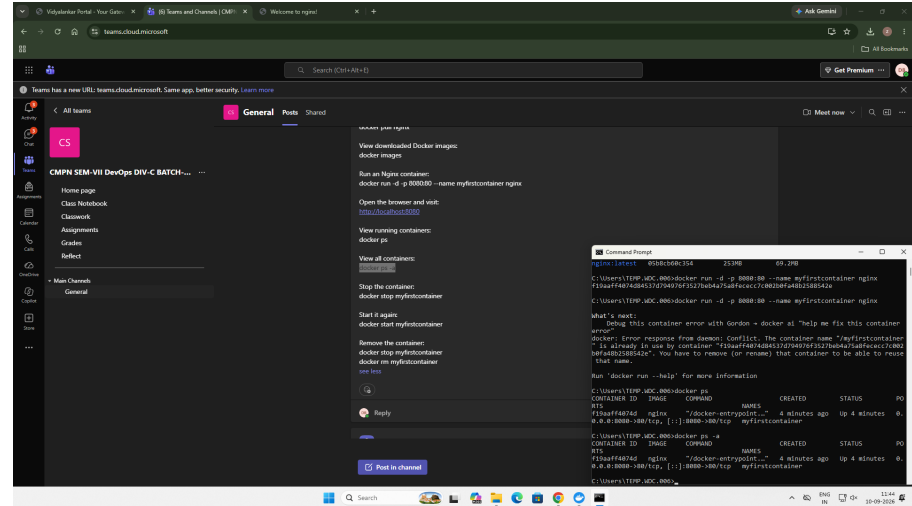


Task 4: View Running Containers

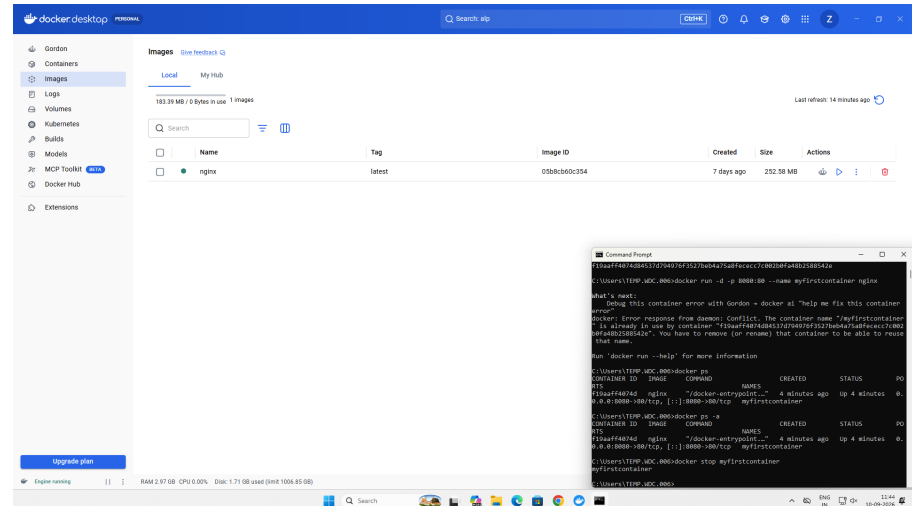


Container Lifecycle

Task 8: View All Containers



Task 8: Stop the Container



Stop and Restart

Task 8: Stop / Start Lifecycle Operations

The screenshot shows the Docker Desktop interface. On the left, the 'Containers' tab is selected. The main area displays a table of containers. The 'myfirstcontainer' is shown with its ID 'f19aaf48274d' and image 'nginx'. The 'intelligent_hawking' container is also listed with ID 'f7c91bee91d' and image 'nginx:latest'. A terminal window on the right shows the following commands and outputs:

```

C:\Users\ITDP_MCC> docker stop myfirstcontainer
myfirstcontainer
C:\Users\ITDP_MCC> docker start myfirstcontainer
myfirstcontainer
  
```

Task 8: Container Stopped State

The screenshot shows the Docker Desktop interface. The 'myfirstcontainer' is now shown with a status of 'Stopped'. The terminal window on the right shows the following command and output:

```

C:\Users\ITDP_MCC> docker stop myfirstcontainer
myfirstcontainer
  
```

Task 19 / Container Cleanup: Remove the Container

The screenshot shows the Docker Desktop application. On the left, a sidebar contains navigation options: Dashboard, Containers, Images, Logs, Volumes, Settings, MCP Toolkit, Docker Hub, and Extensions. The 'Containers' tab is selected, displaying a table of running containers. The table has columns for Name, Container ID, Image, Ports, CPU (%), Last started, and Actions. One container, 'intelligent_hawking', is listed with image 'nginx:latest' and is running. Below the table, there are 'Walkthroughs' for 'Multi-container applications' and an 'Upgrade plan' button. At the bottom, system statistics show 'Engine running', 'RAM 3.00 GB', 'CPU 0.58%', and 'Disk 1.71 GB used (out of 100.85 GB)'. An overlaid terminal window shows the following commands and output:

```

C:\Users\ITP\APC>docker ps
CONTAINER ID   IMAGE      NAMES      CREATED    STATUS    PORTS
f7c91bee91d    nginx:latest  myfirstcontainer  4 minutes ago  up 4 minutes  0.0.0.0:8080->80/tcp

C:\Users\ITP\APC>docker stop myfirstcontainer
myfirstcontainer

C:\Users\ITP\APC>docker start myfirstcontainer
myfirstcontainer

C:\Users\ITP\APC>docker stop myfirstcontainer
myfirstcontainer

C:\Users\ITP\APC>docker rm myfirstcontainer
myfirstcontainer
  
```

Conclusion

The demonstrated Docker workflow verified Docker installation, retrieved the Nginx image, ran an Nginx container with host port 8080 mapped to container port 80, opened the Nginx page at <http://localhost:8080>, inspected running and all containers, and exercised stop, start, and remove operations. These screenshots document the core image and container lifecycle operations described in the DevOps Lab-7 practical.